GALGOTIAS COLLEGE OF ENGINEEING & TECHNOLOGY (GCET)



Master of Computer Applications

Mini Project Review 1

RecallQ: AI-Powered Digital Memory & Intelligent File Retrieval System

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RecallQ

AI-Powered Digital Memory & Intelligent File Retrieval System

“Remember what you need not where you saved it.”

Your Digital Second Brain



An intelligent digital memory system that helps you find information based on what you remember, rather than where you saved it.

Problem Statement

Information Overload: Users store a large number of photos, screenshots, PDFs, documents, and notes.

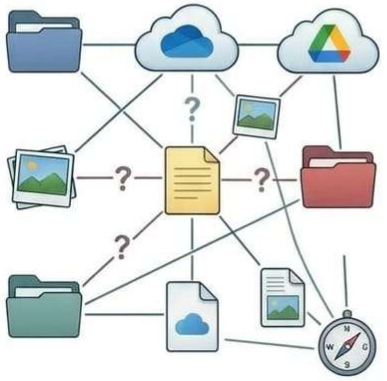
Difficult Retrieval: Finding a specific piece of information becomes difficult when users don't remember the exact file name or location.

Scattered Information: Important information is distributed across different folders and file types.

Time & Frustration: Searching manually takes time and often leads to duplicate files or forgotten information.









"The problem is not storing information — it is remembering where to find it."

OBJECTIVE

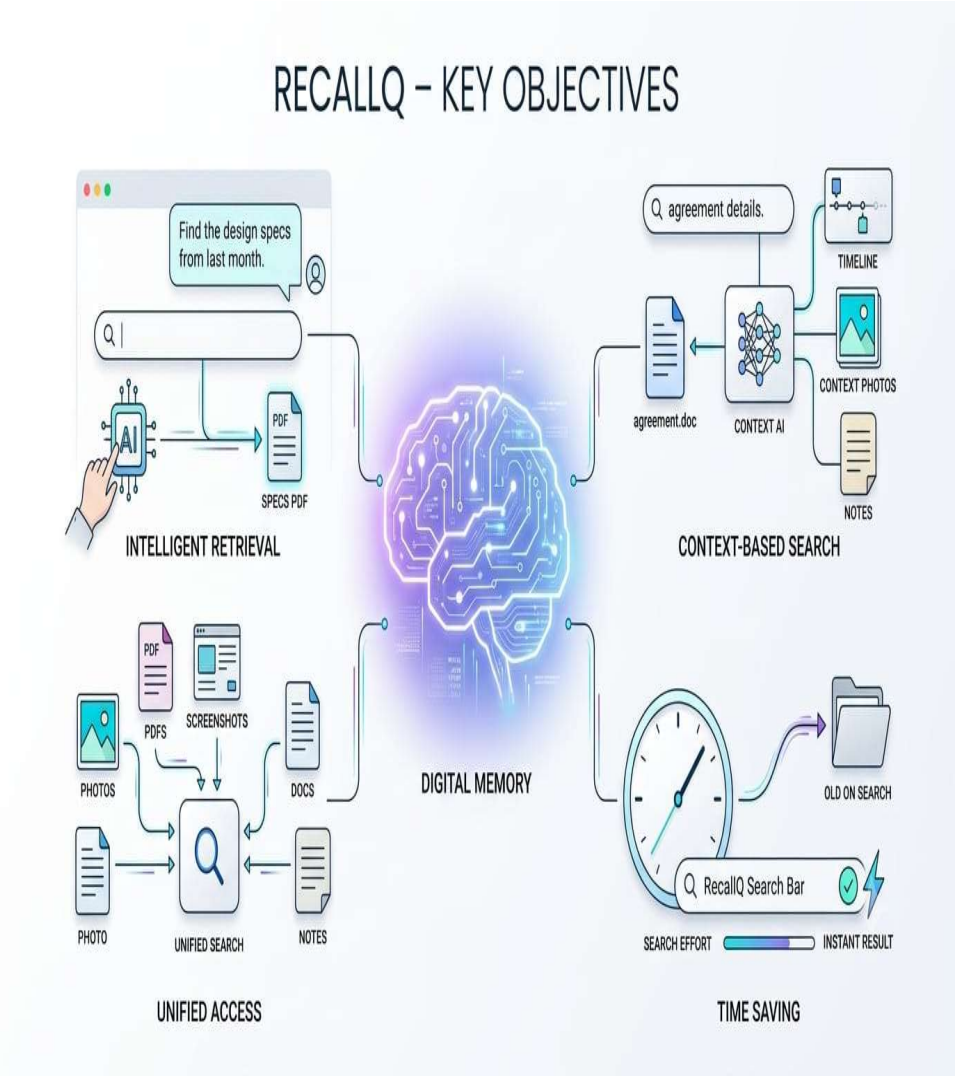
Intelligent Retrieval — Find stored information using natural-language queries.

Context-Based Search — Retrieve relevant files based on meaning and context, not just file names.

Unified Access — Search across photos, screenshots, PDFs, documents, and notes from one place.

Time Saving — Reduce the effort and time required for manual file searching.

Digital Memory — Help users preserve and recall important information when they need it.



Proposed Solution

RecallQ provides an AI-powered digital memory system that understands stored content and retrieves relevant information using natural-language queries.

1. **Collect:**

Bring together **photos, screenshots, PDFs, documents, and notes** in one digital memory.

2. Understand:

AI processes the stored content and identifies its **text, meaning, and context**.

3. Organize:

Information is intelligently indexed so that related content can be connected and retrieved efficiently.

4. Search:

Users describe what they remember using **natural language**, without needing the exact filename or folder.

5. Recall:

RecallQ identifies and presents the **most relevant information or files** to the user.

RecallQ – Workflow

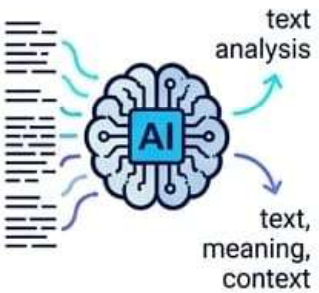
STAGE 1

DIGITAL INFORMATION



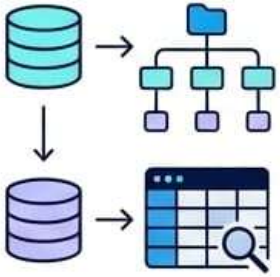
STAGE 2

AI CONTENT UNDERSTANDING



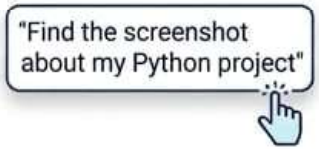
STAGE 3

INTELLIGENT INDEXING



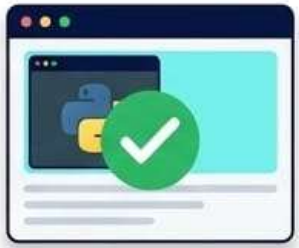
STAGE 4

NATURAL-LANGUAGE QUERY



STAGE 5

RELEVANT INFORMATION RETRIEVED



Stage 1

photos, analyhots, PDFs, & notes

Stage 2

understanding text, meaning, context

Stage 3

organized, searchable digital memory

Stage 4

natural-language query isn input

Stage 5

rerlevant information screenshot

Search by what you remember, not by where you saved it.

Key Features

- **Multi-Format Support**

Supports photos, screenshots, PDFs, documents, and notes in one system.

- **Natural-Language Search**

Allows users to search using everyday language instead of exact file names.

- **Context-Aware Retrieval**

Identifies relevant information based on the meaning and context of the user's query.

- **OCR-Based Text Extraction**

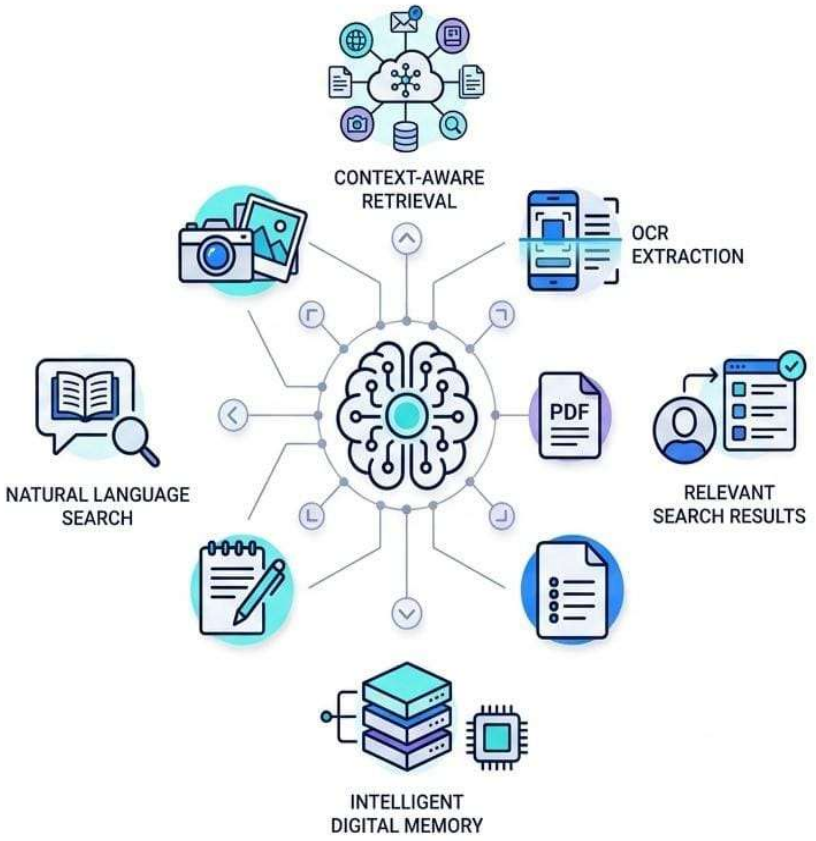
Extracts text from images and screenshots, making their content searchable.

- **Intelligent Digital Memory**

Organizes stored information into a structured and searchable digital memory.

- **Relevant Results**

Retrieves and presents the most relevant files or information for the user's query.



Technology Stack

FRONTEND

HTML — Interface structure
CSS — Styling & layout
JavaScript — User interaction
PyWebView — Desktop interface

BACKEND

Python — Core programming language
Flask — Backend logic & routing

AI & SEARCH

OCR — Text extraction from images
NLP / Text Processing — Content processing
Embeddings — Meaning representation
Semantic Search — Context-based retrieval

DATABASE & STORAGE

SQLite — Metadata & indexing
Local File Storage — Original user files

A lightweight Python-based architecture combining web technologies with AI-powered information retrieval.

System Architecture

1. Information Ingestion

This is **how RecallQ stores and understands your files**:
Files → Extract Content → Understand Content → Store & Index

- **Files** → Photos, screenshots, PDFs, documents, notes
- **Extract Content** → Get text and useful information from files
- **Understand Content** → AI understands the meaning/context
- **Store & Index** → Save information so it can be searched later

2. Information Retrieval

This is **how the user finds something later**:
User Query → Understand Query → Find Match → Show Result

- **User Query** → User asks what they remember
- **Understand Query** → RecallQ understands the request
- **Find Match** → Searches the stored information
- **Show Result** → Displays the relevant file/information

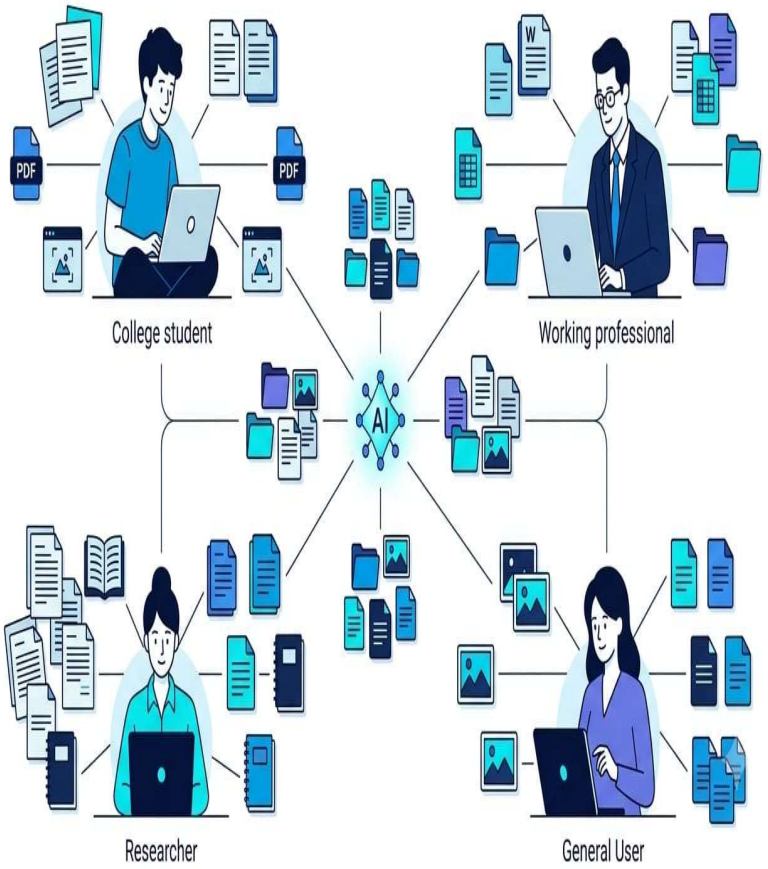
Target User

Students 🎓
Find lecture notes, screenshots, assignments, PDFs and study material.

Working Professionals 💼
Quickly retrieve documents, meeting notes, reports and reference files.

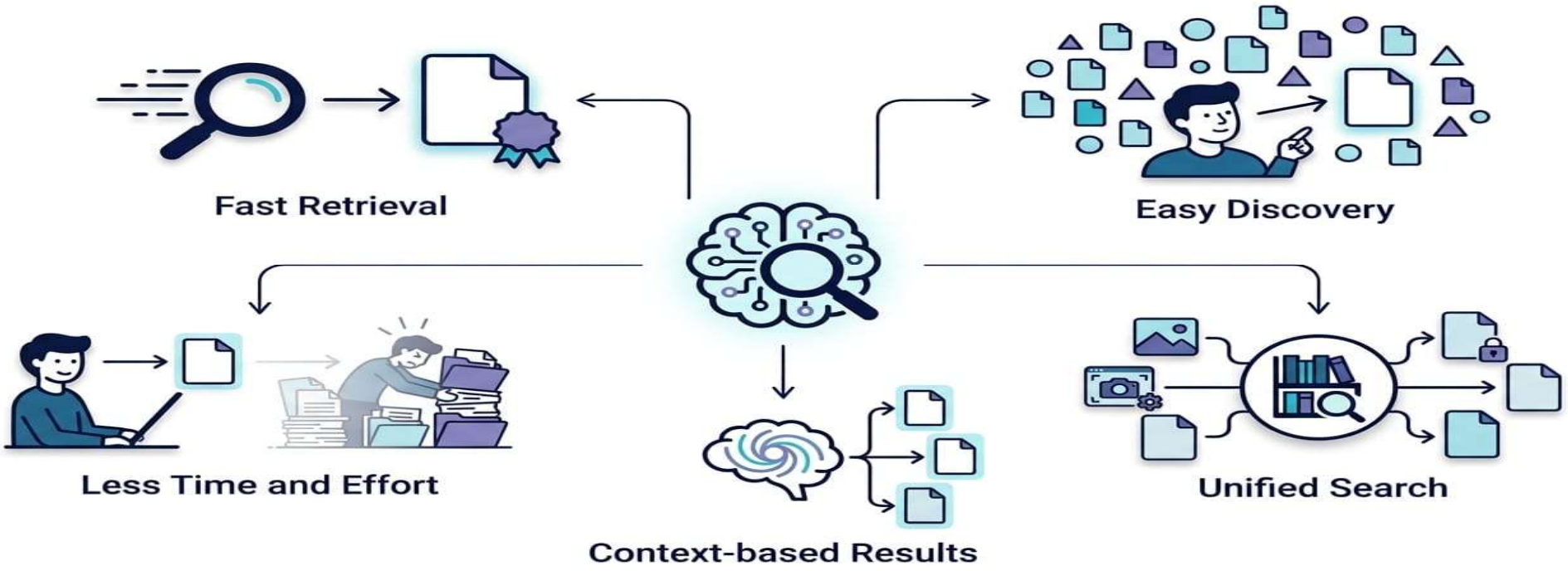
Researchers 📚
Find relevant papers, notes, screenshots and collected information.

General Users 👤
Search personal documents, photos, saved information and important files.



“RecallQ helps users find information by describing what they remember.”

Expected Outcomes & Benefits



“RecallQ makes stored information easier to find, understand, and recall.”

Team Roles & Responsibilities

Nitish Yadav	Payal Kumari	Rahematullah
Frontend & UI Development	Backend, AI & Database	Documentation, Testing & Support
• Design user interface	• Develop backend using Python & Flask	• Prepare project documentation
• Develop pages using HTML, CSS & JavaScript	• Handle application logic & file processing	• Organize project requirements
• Implement file upload & search interface	• Work on OCR & text processing	• Perform functional testing
• Connect frontend with backend	• Manage database & file metadata	• Identify and report errors
• Support testing & integration	• Support semantic search & integration	• Support presentation & project activities

Future Scope

Where RecallQ Can Go Next

Expanding from intelligent file retrieval into a complete digital memory assistant.





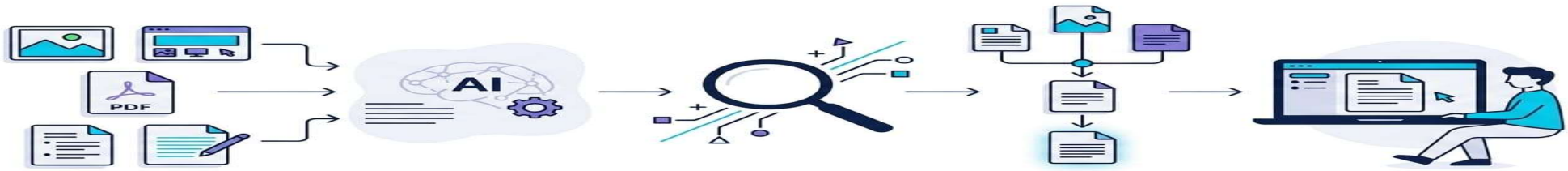
Conclusion

RECALLQ

An AI-powered digital memory and intelligent file retrieval system that helps users find stored information using what they remember, rather than where they saved it.

Key Takeaway

Store → Understand → Search → Recall



“RecallQ makes digital information easier to find, access, and recall.”

THANK YOU

Thank You for Your Attention!!